

Exercise class 11

Partial differential equations

1. Kreyszig Section 12.1, Problems 17, 19, 22.
2. Kreyszig Section 12.3 Problems 5, 15-16.
3. Wave equation with Dirichlet boundary condition and initial condition of zero displacement and constant initial velocity. Solution using Fourier series.
4. Heat equation with insulated ends (Neumann boundary condition) and step function initial condition. (Kreyszig 12.6 example 4)
5. Heat equation with one insulated end (Neumann boundary condition) and one fixed temperature end (Dirichlet) and constant temperature initial condition.